

Learning Neural Control Barrier Functions from Expert Demonstrations using Inverse Constraint Learning

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Motivation

- Control Barrier Functions (CBFs) provide efficient safety filters that minimally correct a nominal controller to maintain safety.
- Training neural CBFs **requires** labelling states as safe/unsafe via the **dynamics-dependent Backward Reachable Set (BRS)**, which is often hard to obtain, especially for high-dimensional systems.
- Our approach:** With the **system dynamics** and a **safety-agnostic reference controller**, we learn a neural CBF directly from **safe expert demonstrations** via **Inverse Constraint Learning (ICL)**, with no BRS labeling required.

Control Barrier Functions

For control-affine system $\dot{x} = f(x) + g(x)u$, a CBF $B: D \rightarrow \mathbb{R}$ certifies that the safe set $\mathcal{C} = \{x \mid B(x) \geq 0\}$ is *forward invariant* under any control satisfying the following:

$$\exists u \in \mathcal{U} : \nabla B(x)(f(x) + g(x)u) + \alpha(B(x)) \geq 0, \forall x \in D.$$

A CBF-QP policy enforces safety by minimally correcting a nominal control π_{ref} :

$$\pi_{\text{CBF-QP}}(x) = \arg \min_{u \in \mathcal{U}} \|u - \pi_{\text{ref}}(x)\|$$

$$\text{s.t. } \nabla B(x)(f(x) + g(x)u) + \alpha(B(x)) \geq 0.$$

Inverse Constraint Learning (ICL)

Let $J(\pi, c)$ and $J(\pi, r)$ denote the expected cumulative constraint cost and reward of policy π , π_E the expert, $\hat{\pi}_E$ the learner, and κ a cost budget. ICL recovers $\hat{c}_\phi$ via:

$$\hat{c}_\phi = \arg \max_c \underbrace{J(\hat{\pi}_E, c) - J(\pi_E, c)}_{\text{outer: find } \hat{c} \text{ separating } \hat{\pi}_E \text{ from } \pi_E},$$

$$\hat{\pi}_E = \arg \max_{\pi} \underbrace{J(\pi, r) \text{ s.t. } J(\pi, \hat{c}_\phi) \leq \kappa}_{\text{inner: update } \hat{\pi}_E \text{ to maximize reward s.t. } \hat{c}}.$$

- The inner-loop policy is replaced by a CBF-QP policy equipped with π_{ref} and B_θ , trained on states labeled by $\hat{c}_\phi$ as safe ($\mathcal{X}_{\text{safe}} = \{x \mid \hat{c}_\phi(x) \leq \delta\}$) or unsafe ($\mathcal{X}_{\text{unsafe}} = \{x \mid \hat{c}_\phi(x) > \delta\}$).
- In general, we want to obtain a policy whose trajectories **match the expert demonstrations**.

Proposed Method: ICL-CBF

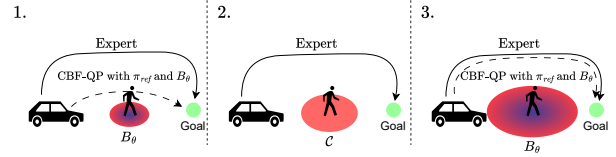


Figure 1. Overview of the ICL-CBF training loop. The constraint $\hat{c}_\phi$ and CBF B_θ are iteratively refined until the learner policy $\hat{\pi}_E$ matches the expert π_E .

- Step 1:** Roll out π_{ref} to collect trajectories; assign safe/unsafe labels via $\hat{c}_\phi$; train B_θ on the labeled states.
- Step 2:** Deploy the CBF-QP policy $\hat{\pi}_E$; update $\hat{c}_\phi$ to maximally separate $\hat{\pi}_E$ from π_E .
- Step 3:** Retrain B_θ on the refined labels; iterate until convergence.

Quantitative Results

Table 1. Success rate (SR) and constraint violation rate (CR) across four environments. L-CBF uses ground-truth labels.

Env.	Met.	iDBF	ROCBF	ICL-CBF	L-CBF
Single integrator	CR	99.2±0.7	0.0	0.0	0.0
	SR	0.8±0.7	9.8±2.1	80.6±4.4	86.2±1.9
Inverted pendulum	CR	2.8±1.2	5.0±1.9	0.2±0.4	0.6±0.8
	SR	97.2±1.2	95.0±1.9	99.8±0.4	99.4±0.8
Dubins car	CR	0.0	69.3±23.3	1.8±1.3	0.3±0.6
	SR	75.0±25.0	6.4±3.4	97.6±1.5	99.6±0.8
Quadrotor	CR	65.7±22.1	75.0±25.0	17.1±6.6	1.5±1.1
	SR	2.8±1.0	0.0	77.2±4.3	98.0±1.3

- ICL-CBF **surpasses both baselines** in success rate (SR) and constraint violation rate (CR) across all four environments.
- ICL-CBF achieves **safety performance comparable to L-CBF** without access to any ground-truth annotations.

Recovered Constraint and CBFs (Single Integrator)

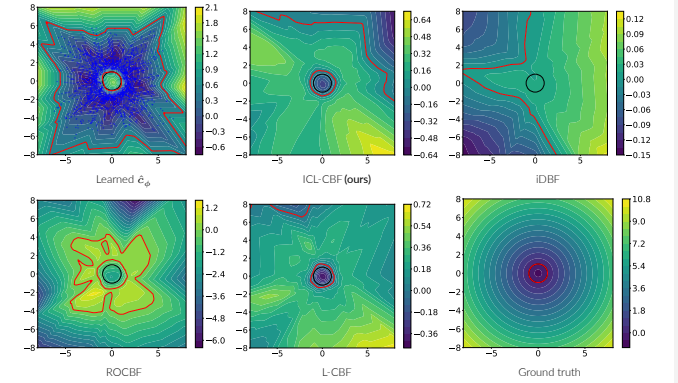


Figure 2. Recovered constraint and CBFs for the single integrator. The red border indicates the zero-level set of the constraint/CBFs; black circle denotes the ground-truth unsafe set.

- The recovered constraint closely approximates the ground-truth safe-set boundary.
- The resulting ICL-CBF labels closely match those of L-CBF, demonstrating that accurate constraint recovery yields **high-quality state annotations** without any ground-truth data.

Sensitivity Analysis

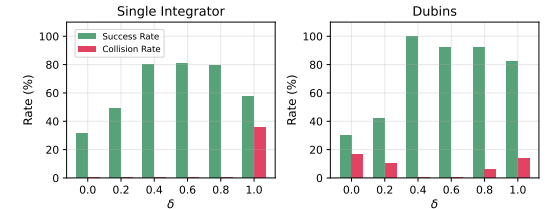


Figure 3. Success rate (SR) and constraint violation rate (CR) as a function of the labeling threshold δ on the single integrator and Dubins car environments.

- The threshold δ requires **no exhaustive tuning**, indicating robustness to hyperparameter selection.